▲ COTTON PRODUCTION
*The first mills were equipped with machines such as the spinning "mule," which spun fiber into thread and sped up the production of cotton. Early mills were set up by fast-flowing rivers to drive the water wheels they used for power. Later mills were powered by steam and located near coalfields. Workers—including children—spent long hours working the machines.*

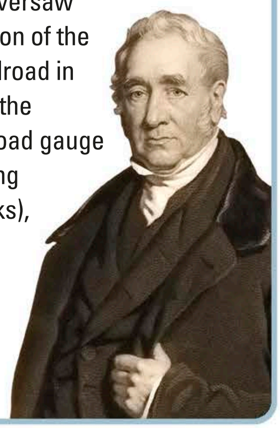
# The Industrial Revolution

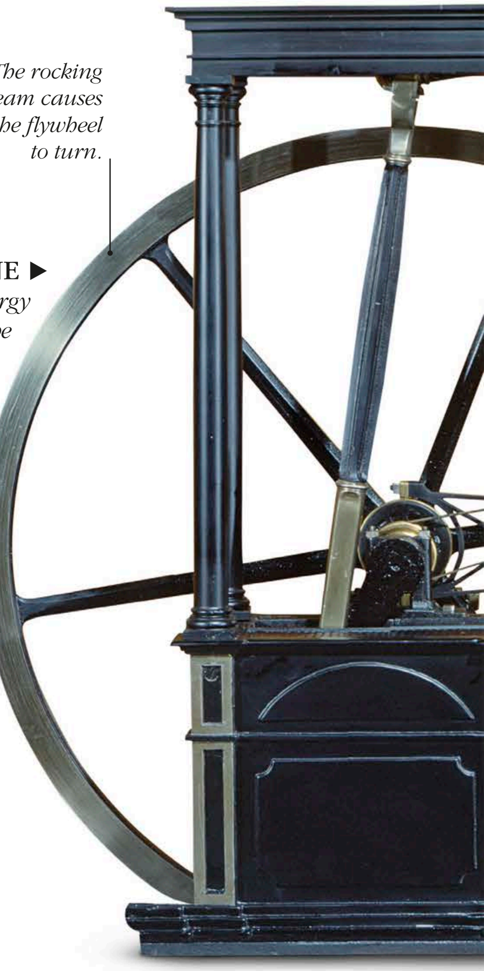
The invention of new technology in mid-18th-century Britain led to a period of rapid development and social change called the Industrial Revolution, which quickly spread around the world. The Industrial Revolution brought about a shift from a traditional agricultural way of life to one in which most people worked in mills, factories, and mines.

The Industrial Revolution originated in mines and in the cotton industry. New inventions such as the spinning jenny and the flying shuttle, which sped up the process of making textiles, made it possible to replace human labor with machines. Within a few decades, steam engines were being used to power machines in large factories. These engines required coal for fuel, so mines were dug deeper to meet the growing demand. Newly constructed railroads and canals carried raw materials and goods around the country, while large ships were built to export them worldwide. Industrialization quickly spread beyond the borders of Britain. By the 1850s, the US and Germany were beginning to overtake Britain in industrial production. The Great Exhibition, held in London in 1851 to showcase industry, contained exhibits not just from Britain but from all over the world.

## GEORGE STEPHENSON

Railroad pioneer George Stephenson (1781–1848) designed his first steam locomotive in 1814 to haul coal wagons. He oversaw the construction of the first public railroad in 1825, devised the standard railroad gauge (a fixed spacing between tracks), and built railroads all over Britain.





*4. The rocking beam causes the flywheel to turn.*

THE STEAM ENGINE ▶
*The first steam engines wasted a lot of energy because the cylinder that powered them had to be repeatedly heated and cooled. James Watt made them more efficient by making sure the cylinder stayed hot continuously. These machines could now be used to drive machinery. Watts patented his engine in 1769 and supplied it to mills and factories.*

MACHINE WRECKERS ▶
*In the 1810s, put out of work by the new textile factories, gangs of traditional hand spinners and weavers began breaking into factories to destroy the machines that had deprived them of their livelihoods. They were called Luddites after their supposed leader, Ned Ludd.*